

AI-BFSI Internship Project Documentation

Empathy Map

Designing an effective software solution requires more than identifying technical requirements. It demands a clear understanding of the people who will use the system, the challenges they experience, the decisions they make and the expectations they have while interacting with a solution. This document develops an empathy map for the target users of the Subscription Waste Detector to ensure that subsequent design and development activities remain focused on genuine user needs.

Purpose of this Document

This document analyses the mindset, behaviours, frustrations, expectations and daily experiences of individuals managing multiple digital subscriptions. The insights obtained through this analysis form the foundation for user-centred requirements gathering, interface design and feature prioritisation throughout the project.

Executive Summary

The Subscription Waste Detector is designed to improve financial awareness by helping users identify and analyse recurring subscription expenses. While the previous document established the business problem, understanding the user behind that problem is equally important. Different individuals manage subscriptions differently, yet many share similar frustrations when attempting to monitor recurring financial commitments.

An empathy map is a user-centred design technique that captures the user's perspective by analysing how they think, feel, communicate and behave while interacting with a particular problem. Rather than focusing on software functionality, it focuses on understanding the human experience behind the problem. This enables software solutions to be designed around real user needs instead of assumptions.

Objectives

- Understand the target user.
- Identify user motivations.
- Recognise recurring frustrations.
- Understand behavioural patterns.
- Support user-centred design decisions.

Document Scope

This document focuses on individuals managing multiple digital subscriptions such as streaming services, productivity software, cloud storage and educational platforms. It analyses user behaviour associated with recurring subscription management and does not cover complete personal finance management.

Expected Outcome

By the end of this document, the project team will have a clear understanding of the target user's behaviours, expectations and pain points, enabling future design decisions to remain aligned with genuine user needs.

1. Target User Profile

The Subscription Waste Detector is intended for individuals who actively use multiple digital subscription services and wish to gain better visibility into recurring financial commitments. Although users differ in age, profession and lifestyle, they exhibit similar behavioural patterns while managing subscriptions. The following profile represents the characteristics commonly observed among the intended user group and serves as the basis for the empathy analysis presented in this document.

Representative User

User Category	Individual Digital Consumer
Technology Usage	Regular user of subscription-based applications and online payment systems.
Financial Behaviour	Maintains multiple recurring subscriptions with different renewal cycles.
Primary Objective	Understand and optimise recurring subscription expenditure.
Decision Style	Prefers simple insights and visual summaries over manual transaction analysis.

User Characteristics

- Uses multiple subscription-based digital services.
- Frequently pays through UPI, cards or digital wallets.
- Reviews financial statements only occasionally.
- Often forgets inactive subscriptions.
- Wants better awareness before subscription renewals.
- Prefers simple dashboards instead of lengthy reports.

Profile Summary

The target user is digitally active and comfortable with online payment systems but experiences difficulty maintaining visibility into recurring subscription expenses. The primary requirement is not additional financial data but a clearer understanding of existing subscription commitments.

2. Empathy Map

The empathy map summarises the user's perspective while managing multiple recurring digital subscriptions. Instead of describing software features, it captures the user's thoughts, emotions, conversations and behaviours associated with recurring subscription management.

Representative User

Digital Consumer

Managing Multiple Recurring Subscriptions

THINK

- Monthly cost
- Unused services
- Upcoming renewals
- Reduce expenses
- Spending visibility

FEEL

- Concerned
- Unsure
- Frustrated
- Overwhelmed
- Financially cautious

SAY

- "I forgot this subscription."
- "Why am I still paying?"
- "I need reminders."
- "Everything together."
- "This should be simpler."

DO

- Checks statements
- Reviews expenses
- Cancels subscriptions late
- Compares spending
- Continues unused subscriptions

Key Insight

The user's greatest challenge is not the absence of transaction data but the absence of a consolidated view of recurring subscription commitments. Improving subscription visibility therefore has a greater impact than simply recording financial transactions.

3. Empathy Analysis

The empathy map provides a concise overview of user behaviour. This section expands those observations by explaining why users think, feel, communicate and behave in a particular manner while managing recurring digital subscriptions. These behavioural insights form the basis for user-centred system design.

Think

Users are primarily concerned with understanding how much they spend on subscriptions each month. They often question whether every active subscription continues to provide value and whether recurring expenses can be reduced without affecting their daily activities. As the number of subscriptions increases, users seek a simpler way to review recurring financial commitments.

Feel

Although users appreciate the convenience of subscription-based services, they frequently experience uncertainty regarding upcoming renewals and inactive subscriptions. This uncertainty gradually reduces financial confidence and creates concern about avoidable recurring expenses.

Behaviour Insight

Users do not lack financial information; they lack clarity. Their thoughts and emotions consistently indicate the need for improved visibility rather than additional transaction records.

4. User Behaviour Analysis

Say

Users frequently express frustration after discovering recurring charges for subscriptions they no longer use. They often mention the need for a single location where all subscriptions can be reviewed and monitored without manually analysing transaction histories.

Do

Most users periodically review bank statements or digital payment history to understand their expenses. Subscription cancellations usually occur only after an unexpected renewal or when recurring charges become noticeable, indicating that current review methods are largely reactive rather than proactive.

Primary Pain Points

Limited Visibility

Difficulty identifying all active subscriptions from transaction records.

Automatic Renewals

Recurring payments continue without regular review.

Fragmented Information

Subscription records remain scattered across multiple payment methods.

Lack of Guidance

Users receive transaction records but limited assistance in optimising subscription spending.

Key Observation

The identified pain points are behavioural rather than technical. Users already possess transaction information but require meaningful analysis that converts raw financial records into understandable subscription insights.

5. User Needs and Design Implications

The empathy analysis identifies several user expectations that directly influence the design of the Subscription Waste Detector. Each implemented feature has been selected to address a specific user need identified during the empathy study.

User Need	Current Implementation
View all subscriptions together.	Recurring subscription detection and subscription listing.
Understand recurring spending.	Monthly spending dashboard and visual analytics.
Receive renewal awareness.	Renewal reminder functionality.
Identify unnecessary subscriptions.	AI-powered recommendations using the Groq API.
Simple financial understanding.	Interactive dashboard with categorized subscription analysis.

Design Implications

- Prioritize clarity over complexity.
- Present information visually.
- Reduce manual effort.
- Support informed financial decisions.
- Improve recurring subscription awareness.

6. Conclusion

This empathy map establishes a comprehensive understanding of the target users of the Subscription Waste Detector. By examining how users think, feel, communicate and behave while managing recurring subscriptions, the document highlights the practical challenges that influence their financial decisions.

The analysis demonstrates that the primary user requirement is improved visibility into recurring subscription commitments rather than additional financial transaction records. These insights have guided the selection of features included in the current implementation while maintaining alignment with the defined project scope.

Transition to Document 04

The behavioural insights obtained from this empathy analysis provide the foundation for the next stage of software engineering. The following document, **Requirements Gathering and Analysis**, converts these user needs into formal functional and non-functional requirements that define the capabilities of the proposed system.